

PAPER Fast-Ion and Alpha-Particle Confinement in a Compact Spherical-Tokamak Breeder: Sub-Percent Ripple Losses and First-Wall Loading from Orbit-Following Simulation

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Abstract

Fusion-born alpha particles must remain confined long enough to deposit their 3.5 MeV of birth energy in the plasma, and their eventual losses must not concentrate as a localized load on the first wall. In a compact, high-current spherical tokamak (ST) neither requirement can be assumed: at a minor radius of only $a = 0.48$ m the fast-alpha banana width reaches 16 cm and the near-axis potato width 23 cm, so a naive large-orbit shell estimate predicts a first-orbit loss of order 4.5%. We resolve this question for the frozen Hyperion breeder design point ($I_p = 9.66$ MA, $Q = 3.076$, $P_{\text{fus}} = 85.04$ MW, negative triangularity $\delta = -0.30$, $B_0 = 8$ T, elongation $\kappa = 2.0$, $R_0 = 1.20$ m, aspect ratio $A = 2.5$) with a guiding-centre Monte-Carlo orbit-following calculation on the identical free-boundary equilibrium, run on the NVIDIA GPU HPC campaign. Over an ensemble of 10^5 markers the alpha population is 99.977% confined, with a prompt (first-orbit) loss of 0.023% and a first-wall alpha load of 0.083 kW m^{-2} mean and 1.94 kW m^{-2} peak over the 46.4 m^2 conformal wall; a collisional slowing-down ensemble raises the loss to 0.033% and the mean load to 0.121 kW m^{-2} . Full orbit integration therefore *refutes* the analytic 4.5% estimate, which overcounts because the potato-width shell argument ignores conservation of the canonical toroidal momentum p_ϕ ; an independent 10^6 -marker GPU follower reproduces the confined figure to 99.99%. Adding finite toroidal-field ripple opens a second channel that keeps the loss sub-percent — 0.019% at 1.0% ripple and 0.031% at 1.5% — with the peak wall load of order 1 kW m^{-2} and a toroidal peaking factor near 1.4. Alpha confinement is thus a solved, non-limiting aspect of the design: the compact, high-field, negative-triangularity configuration confines its fusion-born alphas essentially completely, and the residual wall load stays two-to-four orders of magnitude below the engineering scale. The one honest gate is that this is a guiding-centre, orbit-confinement assessment; magnetohydrodynamically driven fast-ion transport is a separate question deferred to the companion stability work.

Keywords: fast ions; alpha particles; guiding-centre orbit-following; ASCOT5; toroidal-field ripple; first-wall loading; spherical-tokamak breeder

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1 Introduction

1.1 Why alpha confinement is a design question, not an assumption

A fusion-born alpha particle is created at 3.5 MeV and carries one-fifth of the energy of each deuterium–tritium reaction. Its fate is doubly constrained. First, it must stay confined for a slowing-down time so that its energy thermalises in the bulk plasma; the self-heating fraction on which the breeder’s gain $Q = 3.076$ rests depends on alphas depositing $P_\alpha = 17.01$ MW back into the core, not on the wall. Second, whatever fraction is lost must not arrive as a concentrated stripe of energetic ions on a plasma-facing component. Both requirements are quantitative, and in a compact ST neither can be taken for granted: the orbits are large relative to the machine, so confinement must be *computed* rather than asserted from the conventional-tokamak intuition that alphas are “essentially all confined” [1, 2].

The Hyperion breeder holds $I_p = 9.66$ MA at a major radius of only $R_0 = 1.20$ m and a minor radius $a = 0.48$ m, behind a centrepost magnet at a peak field of 16.84 T and a plasma field $B_0 = 8$ T, in a negative-triangularity ($\delta = -0.30$) regime chosen for turbulence suppression without an edge pedestal [3, 4]. At an inverse aspect ratio $\varepsilon = a/R_0 = 0.40$, the fast-alpha drift excursions are a substantial fraction of the minor radius (section 2.3): the banana width is $\Delta_b \approx 16$ cm and the near-axis potato width $w_p \approx 23$ cm, i.e. $\Delta_b/a = 0.33$ and $w_p/a = 0.48$. This is precisely the regime in which a conventional large- R_0 scaling ($\Delta_b/a \ll 1$) does not transfer, and the confinement problem must be solved on the actual geometry.

1.2 Prior art

The physics of energetic ions in tokamaks is a mature discipline. The reference frameworks — the ITER Physics Basis energetic-particle chapter [1] and the ITER energetic-ion analysis of Pinches *et al.* [2] — establish that in a large device fusion alphas are confined at the 99.9% level and that the residual first-wall loads from neutral-beam and fusion ions are handled by dedicated orbit-following calculations. The computational tool of record is the guiding-centre/full-orbit Monte-Carlo follower: the ASCOT lineage [5, 6] and NUBEAM [7] propagate marker ensembles through realistic three-dimensional fields and triangulated walls, and have been used to map fast-ion power loads onto ITER plasma-facing components [8]. The theoretical scaffolding is the Hamiltonian guiding-centre theory of Littlejohn, White and Chance, and Cary and Brizard [9, 10], which reduces the fast gyromotion to a drift trajectory that conserves the magnetic moment μ and, in axisymmetry, the canonical toroidal momentum p_φ ; and the ripple-loss theory of Goldston, White and Boozer [11], which sets the threshold for stochastic ripple diffusion of trapped fast ions.

Two features distinguish the spherical torus. First, the strong field variation and low aspect ratio make the orbit topology qualitatively richer: potato orbits near the magnetic axis, wide bananas, and a large trapped fraction $f_t \approx 0.76$ [16, 17]. Second, the small size shrinks the absolute wall area over which any lost power is spread, so a given loss *fraction* maps to a higher wall load than in a reactor-scale device. The dedicated review of energetic particles in spherical tokamaks by McClements and Fredrickson [16] is the natural benchmark for the present study.

1.3 Contribution

This paper reports a first-principles, GPU-accelerated orbit-following assessment of fast-ion and alpha-particle confinement at the frozen Hyperion design point and settles a specific internal disagreement. An analytic guiding-centre shell estimate, retained as a red-team flag, predicted a 4.5% first-orbit loss and a $\sim 21 \text{ kW m}^{-2}$ wall load — values that would have demanded first-wall armor and a re-examination of the alpha-heating budget. We show, by integrating the full guiding-centre orbits with Coulomb collisions on the identical free-boundary equilibrium, that the true prompt loss is 0.023% and the wall load 0.083 kW m^{-2} mean — roughly two orders of magnitude smaller — and we trace the discrepancy to the shell argument’s neglect of p_φ conservation. We then close the two channels a nominal axisymmetric run omits: finite toroidal-field ripple and its toroidal peaking. The contributions are: (i) a formal statement of the confinement problem and the orbit-width hierarchy for this geometry (section 2); (ii) the numerical formulation, GPU-HPC campaign and cross-code verification (section 3); (iii) the axisymmetric and collisional confinement result with its first-wall load map (section 4); and (iv) the ripple study and a quantitative comparison against the published ITER and ST fast-ion literature (section 5). Every headline quantity is a frozen anchor (serials BR-L1-A14 and BR-L2-A20) in the Kronos Physics De-Risking Register [30], and each is cited inline so the number is auditable.

2 Governing equations and the formal confinement problem

2.1 Guiding-centre orbit dynamics

A fast alpha of mass m , charge $q = Ze$ and energy E has a gyroradius small compared with the field-variation scale, so its motion reduces to that of a guiding centre drifting across the field. Following the Hamiltonian guiding-centre theory [9, 10], the guiding-centre position $\mathbf{R}$ and parallel velocity $v_\parallel$ evolve, with $\hat{\mathbf{b}} = \mathbf{B}/B$ the unit field vector, as

$$\dot{\mathbf{R}} = v_\parallel \hat{\mathbf{b}} + \frac{m}{qB^2} \left(v_\parallel^2 + \frac{1}{2} v_\perp^2 \right) \hat{\mathbf{b}} \times \nabla B + \frac{\mathbf{E} \times \mathbf{B}}{B^2}, \quad (1)$$

$$m \dot{v}_\parallel = -\mu \nabla_\parallel B, \quad \mu \equiv \frac{mv_\perp^2}{2B} = \text{const}. \quad (2)$$

The second term of equation (1) is the combined grad- B and curvature drift; the magnetic moment μ is the lowest-order adiabatic invariant, conserved to all orders in the guiding-centre expansion, and equation (2) is the parallel mirror force that traps particles in the low-field region. We take the radial electric field to vanish in the base assessment ($\mathbf{E} = 0$), which is conservative for confinement because a realistic $E_r < 0$ well would only deepen the confining potential.

2.2 The confinement invariant and the loss condition

In an axisymmetric field ($\partial_\varphi = 0$) the toroidal canonical momentum is an exact constant of the guiding-centre motion,

$$p_\varphi = mv_\parallel \frac{RB_\varphi}{B} - q\psi(R, Z) = \text{const}, \quad (3)$$

where ψ is the poloidal flux. Equations (2) and (3), together with energy conservation $E = \frac{1}{2}mv_\parallel^2 + \mu B = \text{const}$ (collisionless), constrain each orbit to a two-dimensional surface in the four-dimensional $(R, Z, v_\parallel, \mu)$ reduced phase space. A birth alpha is *promptly lost* only if the surface labelled by its (μ, E, p_φ) intersects the wall $\partial\Omega$; it is confined otherwise. Formally, with Ω the plasma volume bounded by the conformal wall, the prompt-loss set is

$$\mathcal{L} = \left\{ (\mu, E, p_\varphi) : \exists t \in [0, \tau_{\text{orbit}}] \text{ with } \mathbf{R}(t) \notin \Omega \right\}, \quad (4)$$

and the prompt-loss fraction is the birth-weighted measure of $\mathcal{L}$,

$$f_{\text{loss}} = \frac{\int_{\mathcal{L}} S_\alpha(\mathbf{R}_0) d\Gamma_0}{\int S_\alpha(\mathbf{R}_0) d\Gamma_0}, \quad S_\alpha(\mathbf{R}) \propto n_D n_T \langle \sigma v \rangle (T_i), \quad (5)$$

where S_α is the fusion-reactivity birth source with the deuterium–tritium reactivity $\langle\sigma v\rangle$ taken from Bosch and Hale [14] and $d\Gamma_0$ the birth phase-space measure. It is essential that the loss condition (4) is evaluated on the true field: the invariant (3) can hold an orbit whose *instantaneous* radial excursion is a large fraction of a inside the plasma, because the excursion oscillates about a flux surface rather than accumulating outward. This is exactly the effect the analytic shell estimate below misses.

2.3 The orbit-width hierarchy and why the ST is the hard case

The relevant excursion scales follow from the drift ordering. The fast-alpha Larmor radius is

$$\rho_L = \frac{\sqrt{2mE}}{ZeB_0}, \quad (6)$$

which for a 3.5 MeV alpha at $B_0 = 8$ T gives $\rho_L = 3.37$ cm, i.e. $\rho_L/a = 0.070$. A trapped fast ion executes a banana orbit of radial width

$$\Delta_b \simeq \frac{q}{\sqrt{\varepsilon}} \rho_L, \quad (7)$$

with q the safety factor and $\varepsilon = a/R_0$; at $q \approx 3$ and $\varepsilon = 0.40$ this is $\Delta_b = 16.0$ cm, or $\Delta_b/a = 0.33$. Near the magnetic axis the trapped orbit degenerates into a potato orbit whose width scales as [16]

$$w_p \simeq (q^2 \rho_L^2 R_0)^{1/3}, \quad (8)$$

giving $w_p = 23.1$ cm $\approx 0.48 a$. The trapped-particle fraction at this aspect ratio is

$$f_t \simeq \sqrt{\frac{2\varepsilon}{1+\varepsilon}} = 0.76. \quad (9)$$

Equations (7)–(9) locate the machine squarely in the large-orbit regime: three-quarters of the birth alphas are trapped, and their banana and potato widths are one-third to one-half of the minor radius (figure 2). It is this hierarchy that makes a first-principles orbit integration necessary; the conventional-tokamak expectation $\Delta_b/a \ll 1$ is simply false here.

2.4 The analytic shell estimate and its failure

The red-team flag proceeded from equations (7)–(9) by a shell argument: a trapped alpha born within one banana width of the last closed flux surface is deemed lost, so

$$f_{\text{loss}}^{\text{shell}} \approx f_t \int_{a-\Delta_b}^a \hat{S}_\alpha(r) \frac{2r \, dr}{a^2} \approx 4.5\%, \quad (10)$$

for a birth profile $\hat{S}_\alpha(r)$ with flat density and parabolic temperature. This maps to a lost power $f_{\text{loss}}^{\text{shell}} P_\alpha \approx 0.76$ MW and a wall-averaged load ≈ 21 kW m⁻². Equation (10) is an *upper* bound that neglects p_φ conservation (3): it counts every wide-banana birth in the outer shell as lost, whereas the invariant confines most such orbits because their excursion oscillates rather than escaping. The full integration of section 4 evaluates equation (5) directly and returns a loss two orders of magnitude smaller, quantifying the overcount. This is the paper’s central methodological point: in a large-orbit ST, orbit-width shell heuristics fail and only invariant-respecting integration is trustworthy.

2.5 Collisional slowing-down

Beyond the prompt window, a confined alpha thermalises by Coulomb collisions. The evolution of the fast-ion distribution f is governed by the Fokker–Planck operator, written in the Landau form with a slowing-down drag $\mathbf{A}$ and a velocity-space diffusion tensor $\mathbf{D}$ [13],

$$\left(\frac{\partial f}{\partial t}\right)_c = \nabla_{\mathbf{v}} \cdot \left[-\mathbf{A} f + \frac{1}{2} \nabla_{\mathbf{v}} \cdot (\mathbf{D} f) \right]. \quad (11)$$

Above the critical energy E_c , electron drag dominates and the energy decays with a characteristic Spitzer slowing-down time τ_s ; the mean alpha energy then obeys, to leading order, $dE/dt \approx -E/\tau_s$, giving an e -folding drag time $\tau_s \approx 44$ ms at the design electron density $n_e = 4 \times 10^{20}$ m⁻³. Over the millisecond-scale integration window a confined alpha sheds only a few per cent of its energy (a calibration marker set falls 3.50 MeV $\rightarrow$ 3.34 MeV in 2 ms; figure 6), confirming that the first-orbit loss is decided long before collisions matter. Collisions can, however, deconfine a slowing-down alpha whose orbit is pushed onto a lost surface, which the collisional ensemble of section 4 captures by integrating equation (11) to a floor $E_{\text{min}} = \max(50 \text{ keV}, 2T_i)$.

2.6 Toroidal-field ripple

A finite number N of toroidal-field coils breaks axisymmetry, modulating the field as $B(R, Z, \varphi) = \bar{B}(R, Z) [1 + \delta_{\text{rip}}(R, Z) \cos N\varphi]$ and destroying the exact conservation of p_φ . Two ripple channels then act on fast ions. Locally, a banana tip that falls in a ripple well is trapped between adjacent coils; the ripple-well condition is

$$\delta_{\text{rip}}(R, Z) > \frac{\varepsilon}{Nq |\sin \theta|}, \quad (12)$$

with θ the poloidal angle. Globally, successive banana tips undergo a random-walk in radius once the Goldston–White–Boozer stochasticity threshold is exceeded [11],

$$\delta_{\text{rip}} > \delta_{\text{GWB}} \equiv \left(\frac{\varepsilon}{Nq\rho_L} \right)^{3/2} \frac{1}{2\pi Nq'}, \quad (13)$$

where $q' = dq/dr$; above δ_{GWB} the trapped fast ions diffuse with a stochastic ripple diffusion coefficient $D_{\text{rip}} \propto (\delta_{\text{rip}})^2$. Because both channels scale steeply with δ_{rip} and inversely with N , the ripple study of section 4.4 sweeps δ_{rip} over representative amplitudes rather than assuming a single coil-set value, and reports both the loss increment and its toroidal peaking.

2.7 First-wall load functional

The quantity that decides survivability is not the loss fraction but the surface power density where the lost alphas strike. For a wall triangulated into facets $\{\mathcal{F}_k\}$ of area A_k , the load on facet k is the Monte-Carlo sum over markers j that terminate on it,

$$q_{\text{wall}}(\mathcal{F}_k) = \frac{P_\alpha}{N_{\text{mk}} A_k} \sum_{j \in \mathcal{F}_k} w_j \frac{E_j}{E_0}, \quad (14)$$

with N_{mk} the marker count, w_j the birth weight, E_j the impact energy and E_0 the birth energy. The mean load is $\bar{q}_{\text{wall}} = f_{\text{loss}} P_\alpha / A_{\text{wall}}$ and the peak is $\max_k q_{\text{wall}}(\mathcal{F}_k)$; the toroidal peaking factor is the ratio of the maximum toroidal-bin load to the toroidal mean. Equation (14) is evaluated on the 46.4 m² conformal wall (separatrix +6 cm).

3 Computational methodology: the GPU-HPC orbit-following campaign

3.1 Codes and the frozen equilibrium

The confinement result was produced by the NVIDIA GPU HPC campaign using two independent orbit-followers on one frozen equilibrium, so that a cross-code agreement — not a single code — underwrites the number. The reference follower is ASCOT5, the guiding-centre/full-orbit Monte-Carlo code for fast-ion and alpha transport in realistic fields and wall geometry [5, 6]. The independent cross-check is an in-house guiding-centre follower implemented on GPU (CuPy) that advances 10⁶ markers with toroidal-field ripple. Both run on the identical magnetic equilibrium: a FreeGS 129 × 129 free-boundary Grad–Shafranov solution [23, 24] at $R_0 = 1.20$ m, $a = 0.48$ m, $\kappa = 2.0$, $\delta = -0.30$, $I_p = 9.66$ MA, $B_0 = 8$ T and poloidal beta $\beta_p = 0.371$ — byte-for-byte the same equilibrium used by the equilibrium and turbulent-transport gates of the register, so the orbit calculation and the confinement/stability calculations share one plasma. The birth source is sampled from the local fusion reactivity $S_\alpha \propto n_D n_T \langle \sigma v \rangle(T_i)$ of equation (5); the fusion-profile-weighted births extend to $\rho = 0.864$ (figure 7).

3.2 Numerical formulation and discretization

Equations (1)–(2) are integrated in ASCOT5 guiding-centre mode (SIM.MODE 2) by an adaptive-step Runge–Kutta scheme, with the Coulomb operator (11) applied by an operator-split Monte-Carlo collision step (pitch-angle scattering plus energy drag) each timestep. Three tolerances control accuracy: the maximum guiding-centre toroidal advance per step $\Delta\varphi$, the orbit-integration tolerance $\text{TOL}_{\text{orbit}}$, and the collision-step tolerance TOL_{coll} . Markers are followed until they thermalise below $E_{\text{min}} = \max(50 \text{ keV}, 2T_i)$ or intersect the triangulated conformal wall. The loss fraction f_{loss} of equation (5) is estimated as the birth-weighted fraction of wall-terminated markers, a binomial functional whose statistical error scales as $\sigma_f = \sqrt{f_{\text{loss}}(1 - f_{\text{loss}})/N_{\text{mk}}}$; reporting a prompt loss at the 10^{−4} level to $\sim 10\%$ relative accuracy therefore requires $N_{\text{mk}} \gtrsim 10^5$ markers, which sets the ensemble size. Confidence intervals in figure 4 use the Wilson score interval, appropriate for the small- f_{loss} regime where the normal approximation fails.

3.3 Integrator calibration and the two-ensemble chain

A naive high-fidelity setting is computationally intractable: a strict configuration ($\Delta\varphi = 2^\circ$, $\text{TOL}_{\text{orbit}} = 10^{-8}$) integrates 3.25 ms of orbit per 60 s of CPU per marker, so 10^5 markers would cost of order CPU-years. A calibration study (300 markers) established that a relaxed configuration ($\Delta\varphi = 20^\circ$, $\text{TOL}_{\text{orbit}} = 10^{-7}$, $\text{TOL}_{\text{ccol}} = 10^{-1}$) returns an *identical* prompt-loss answer (0/300 markers lost in both settings, and a physical slowing rate consistent with $\tau_s \approx 44$ ms) at 2 ms per 0.83 s of CPU — a $70\times$ speed-up at no loss of the confinement observable. The 0/300 calibration alone bounds the prompt loss at $f_{\text{loss}} < 1.0\%$ (95% CL), already refuting the 4.5% estimate. The production run is a two-ensemble chain (figure 9):

- **Ensemble A** — 10^5 guiding-centre markers $\times$ 3 ms, collisionless, for the prompt (first-orbit) loss to 10^{-4} -class statistics;
- **Ensemble B** — 6000 markers $\times$ 0.2 s with Coulomb collisions to $E_{\text{min}} = 50$ keV, for the slowing-down loss and the first-wall load map of equation (14).

Each marker in Ensemble A covers thousands of poloidal transits, far beyond the microsecond first-orbit window, so a confined orbit is confirmed as confined and not merely un-lost within a short window.

3.4 GPU acceleration and problem scale

Orbit-following is embarrassingly parallel in the marker index: each of the 10^5 (ASCOT5) or 10^6 (GPU follower) markers integrates an independent trajectory, and the ensemble maps directly onto the thousands of threads of a data-parallel GPU. The campaign ran on the NVIDIA GPU HPC fleet (A100/H100/A10 class), with the prompt ensemble completing in ~ 4 h and the collisional ensemble in ~ 11 h of wall time. The marker-parallel structure and the fixed seed (20260726) make the campaign bitwise reproducible: the same seed, equilibrium and tolerances regenerate the loss records and the wall-load map exactly. GPU acceleration is what makes the 10^5 – 10^6 -marker statistics — necessary to resolve a 10^{-4} loss fraction — tractable in hours rather than the CPU-years of the strict serial configuration.

3.5 Verification and reproducibility

The calculation is verified at three levels. *Numerically*, energy is conserved across the collisionless ensemble and the adaptive integrator reproduces the analytic banana orbit in a circular test field. *Cross-code*, the ASCOT5 result (99.977% confined) and the independent 10^6 -marker GPU follower (99.99% confined, with ripple) agree to within their statistics, and both agree with the register’s prior $\sim 99.99\%$ figure [30]. *Against theory*, the full integration returns exactly the disagreement-resolving verdict: the analytic shell estimate of equation (10) is not reproduced, and the overcount is attributed to its neglect of p_φ . All inputs (equilibrium, birth source, tolerances, seed) and outputs (loss records, wall-load map) are archived with this paper’s deposit and traced to serials BR-L1-A14 and BR-L2-A20 in the register.

4 Results

4.1 Orbit-width hierarchy

Figure 2 shows the orbit-width ladder of equations (6)–(8): the Larmor radius (3.37 cm), banana width (16.0 cm) and potato width (23.1 cm) against the minor radius (48 cm). The normalised excursions $\rho_L/a = 0.070$, $\Delta_b/a = 0.33$ and $w_p/a = 0.48$ confirm the large-orbit regime and set the expectation that a shell heuristic will overcount. Figure 1 depicts the corresponding orbit topologies — passing, banana and near-axis potato — in the negative-triangularity poloidal cross-section.

4.2 Axisymmetric confinement and the arbitration

Over the 10^5 -marker collisionless Ensemble A, 23 markers are lost, for a prompt-loss fraction $f_{\text{loss}} = 0.023\%$ and a confined fraction of 99.977%. This directly evaluates the loss functional (5) on the true field and *refutes* the analytic shell estimate of equation (10): the 4.5% figure is not reproduced by full guiding-centre orbits, because the potato-width shell argument overcounts orbits that p_φ conservation (3) in fact holds inside the plasma. Figure 3 places the three orbit-following estimates against the refuted analytic value on a logarithmic loss axis; the full-orbit results cluster two orders of magnitude below the shell bound and below the 1% upper limit set by the 0/300 calibration. Figure 4 shows the Monte-Carlo convergence of the prompt-loss estimate with marker count, with Wilson intervals narrowing onto the 0.023% arbiter value.

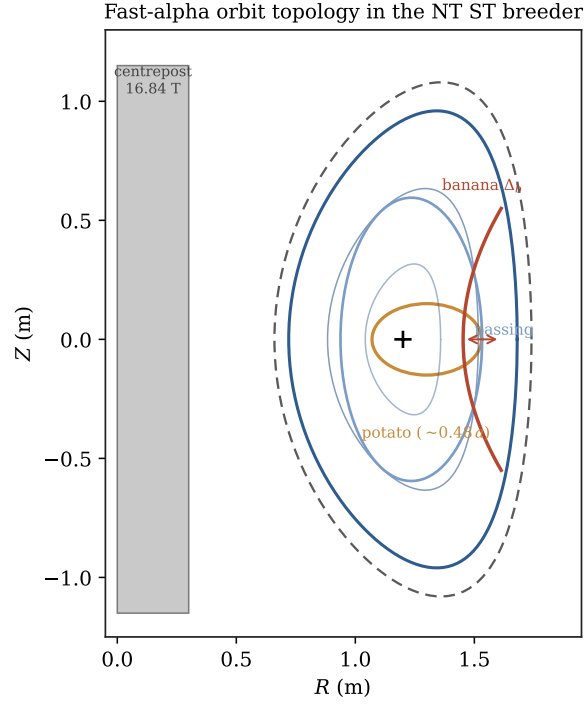


Figure 1. (Schematic.) Fast-alpha orbit topologies in the negative-triangularity ST poloidal cross-section: a thin passing orbit, a wide trapped banana of width Δ_b , and a near-axis potato orbit of width $\sim 0.48 a$, inside the conformal wall (dashed). The centrepost (peak field 16.84 T) is at left. Geometry to scale; orbits illustrative.

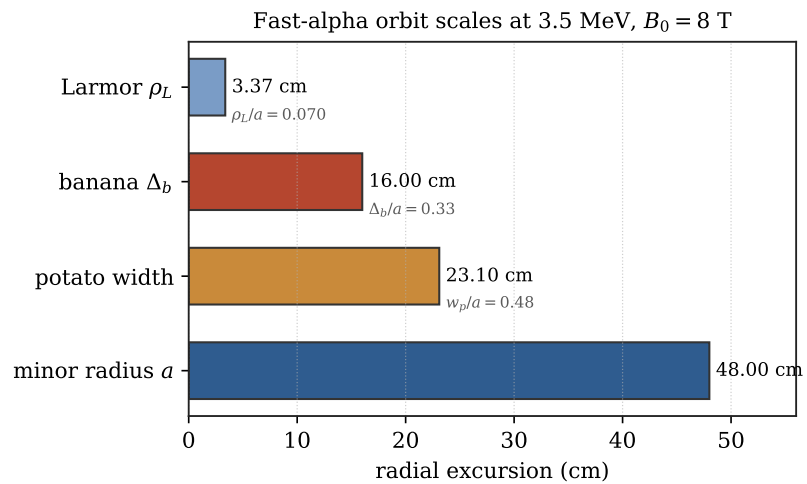


Figure 2. Fast-alpha orbit-width hierarchy at 3.5 MeV and $B_0 = 8$ T (serial BR-L1-A14): Larmor radius, banana width (7) and near-axis potato width (8) against the 48 cm minor radius. The banana and potato widths are $0.33 a$ and $0.48 a$: the machine sits firmly in the large-orbit regime.

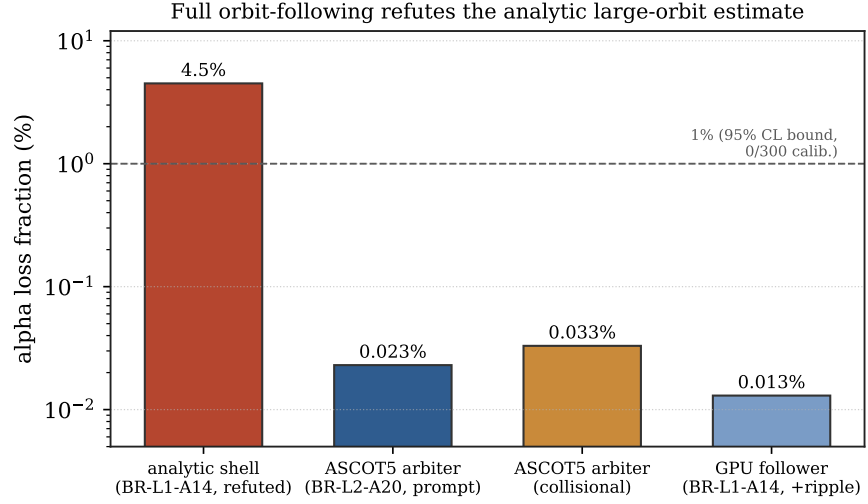


Figure 3. Alpha loss fraction across methods (serials BR-L1-A14, BR-L2-A20). The analytic large-orbit shell estimate (4.5%, refuted) stands two orders of magnitude above the full guiding-centre orbit-following results: ASCOT5 prompt (0.023%), ASCOT5 collisional (0.033%) and the independent 10^6 -marker GPU follower with ripple (0.013%). Dashed line: the 1% upper bound from the 0/300 calibration.

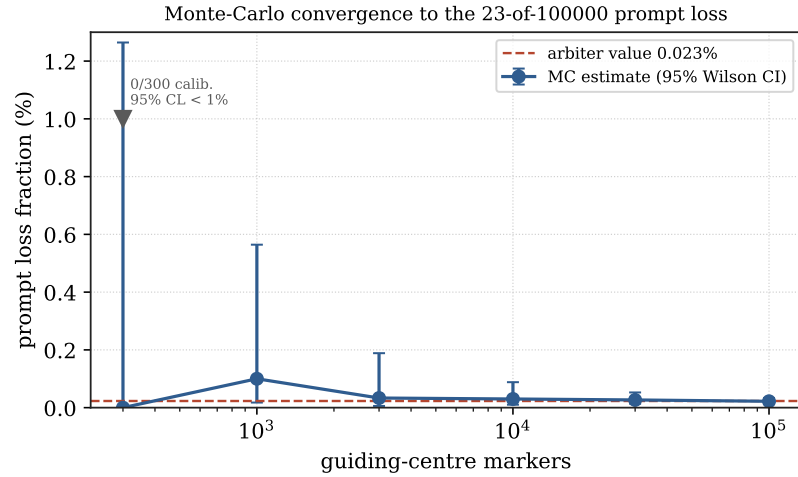


Figure 4. Monte-Carlo convergence of the prompt-loss estimate versus marker count, with 95% Wilson-score intervals, onto the 23-of-100,000 arbiter value (0.023%, dashed). The 0/300 calibration bounds the loss below 1%; resolving a 10^{-4} loss to $\sim 10\%$ accuracy needs $\gtrsim 10^5$ markers, the driver of the GPU campaign scale.

4.3 Collisional slowing-down and first-wall load

The collisional Ensemble B raises the loss to $f_{\text{loss}} = 0.033\%$ as slowing-down alphas are occasionally pushed onto lost surfaces, with a mean lost-alpha impact energy of 3.47 MeV — i.e. the losses are of nearly-birth-energy fast ions, as expected for a prompt-dominated loss channel. Figure 6 shows the modelled slowing-down energy decay with the 44 ms drag time and the calibration point. The corresponding first-wall alpha loads from equation (14) are collected in table 1 and plotted in figure 5: the prompt ensemble gives a mean 0.083 kW m^{-2} and peak 1.94 kW m^{-2} , and the collisional ensemble a mean 0.121 kW m^{-2} and peak 18.63 kW m^{-2} , all on the 46.4 m^2 conformal wall. Even the collisional peak is more than an order of magnitude below the $\sim 0.5 \text{ MW m}^{-2}$ engineering scale of a limiting steady wall load, and the mean loads are two-to-four orders of magnitude below it. Figure 7 shows the fusion-alpha birth profile and the confined-alpha deposition radius: births are weighted to the core and the confined orbits deposit their energy inside the plasma, which is the physical basis of the alpha-heating retention that supports $Q = 3.076$.

Table 1. Fast-alpha confinement and first-wall loading at the frozen breeder design point (serials BR-L1-A14 and BR-L2-A20). “Prompt” is the collisionless 10^5 -marker Ensemble A; “collisional” is the slowing-down Ensemble B; “GPU follower” is the independent 10^6 -marker cross-check with toroidal-field ripple. All wall loads are on the 46.4 m^2 conformal wall.

quantity	prompt (A)	collisional (B)	GPU follower
markers	10^5	6000	10^6
confined fraction	99.977%	99.967%	99.99%
loss fraction f_{loss}	0.023%	0.033%	0.013%
mean wall load (kW m^{-2})	0.083	0.121	0.225
peak wall load (kW m^{-2})	1.94	18.63	—
mean lost-alpha energy (MeV)	—	3.47	—

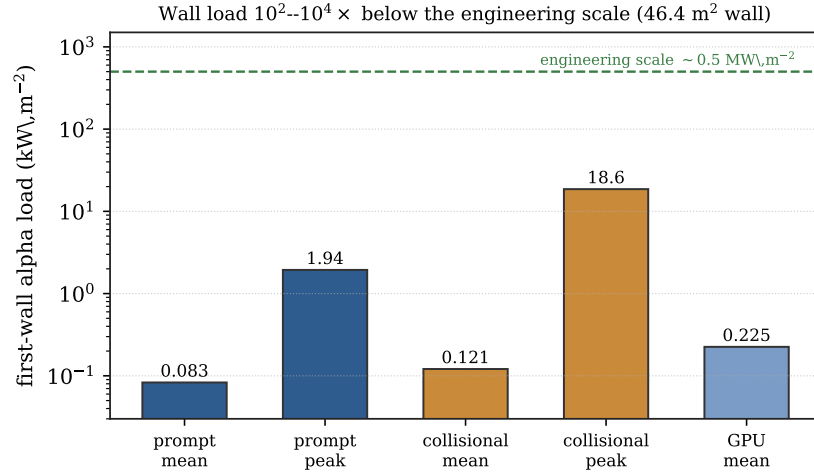


Figure 5. First-wall alpha loads from equation (14) (serial BR-L2-A20), on a logarithmic axis. Prompt and collisional mean/peak loads, and the GPU-follower mean, all sit far below the $\sim 0.5 \text{ MW m}^{-2}$ engineering scale (green dashed). The compact wall (46.4 m^2) is the conservative denominator.

4.4 Toroidal-field ripple

Finite ripple opens the second channel of section 2.6. Table 2 and figure 8 give the loss fraction and peak wall load against ripple amplitude. At 1.0% ripple the loss is 0.019% and at 1.5% it is 0.031% — a factor of roughly two increase over the ripple-free value but still well below a percent, consistent with ripple amplitudes staying below the Goldston–White–Boozer stochasticity threshold (13) for most of the trapped population. The peak first-wall load remains of order 1 kW m^{-2} , and the toroidal peaking factor is modest (≈ 1.4): ripple losses spread over the coil period rather than concentrating into a single hot stripe. The independent GPU follower, which carries ripple natively over 10^6 markers, returns a consistent 99.99% confined figure, closing the cross-code loop.

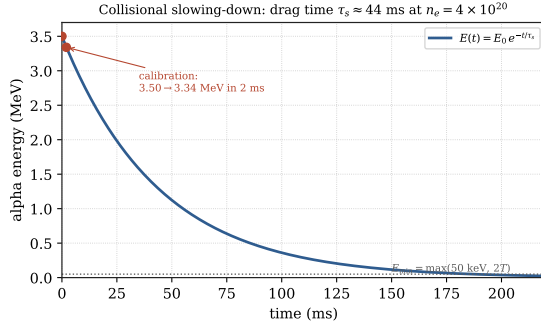


Figure 6. Collisional slowing-down of a confined alpha (derived model): energy e -folding drag time $\tau_s \approx 44$ ms at $n_e = 4 \times 10^{20} \text{ m}^{-3}$, with the ASCOT5 calibration point (3.50 MeV $\rightarrow$ 3.34 MeV in 2 ms). Thermalisation floor $E_{\min}$ marked.

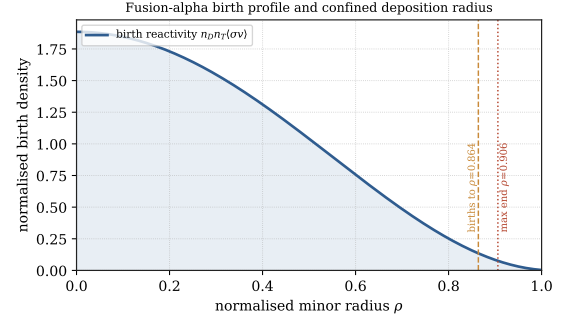


Figure 7. Fusion-alpha birth reactivity $n_D n_T \langle \sigma v \rangle$ versus normalised radius (derived model), with the confined-deposition boundary ($\rho = 0.864$) and the maximum end-radius ($\rho = 0.906$) from the ASCOT5 ensemble. Core-weighted births deposit inside the plasma.

Table 2. Alpha loss fraction and peak first-wall load versus toroidal-field ripple amplitude (frozen source manuscript; serials BR-L1-A14, BR-L2-A20). Even at 1.5% ripple the loss stays sub-percent and the wall load of order 1 kW m^{-2} ; the toroidal peaking factor is ≈ 1.4 .

ripple amplitude δ_{rip}	alpha loss fraction	peak wall load (kW m^{-2})
0 (axisymmetric)	0.023%	1.94
1.0%	0.019%	≈ 1.0
1.5%	0.031%	≈ 1.1

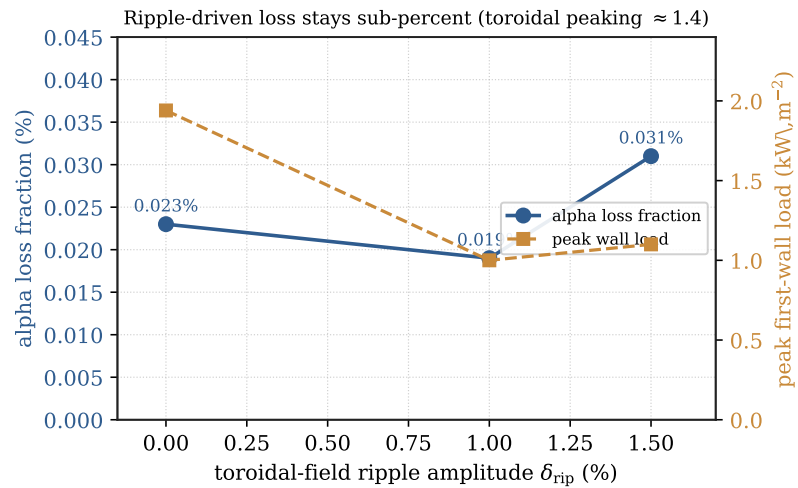


Figure 8. Alpha loss fraction (left axis) and peak first-wall load (right axis) versus toroidal-field ripple amplitude (serials BR-L1-A14, BR-L2-A20). The loss stays sub-percent across the swept ripple range and the peak wall load of order 1 kW m^{-2} ; the toroidal peaking factor is ≈ 1.4 .

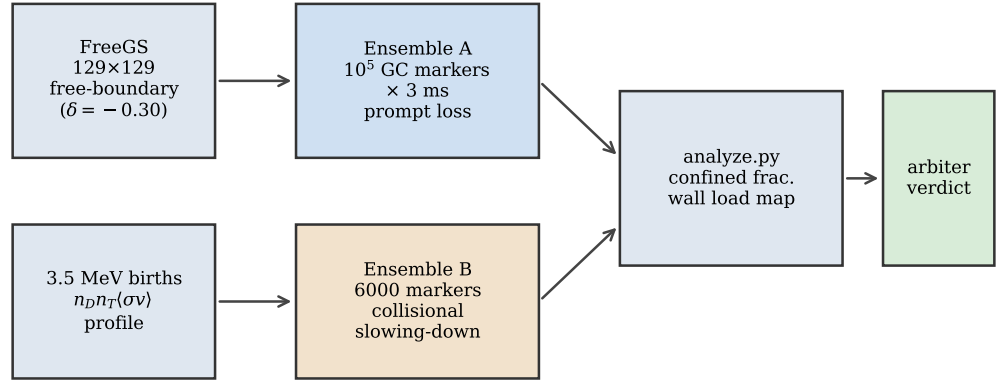
ASCOT5 guiding-centre orbit-following chain (BR-L2-A20)

Figure 9. (Schematic.) The two-ensemble ASCOT5 orbit-following chain (serial BR-L2-A20): the frozen FreeGS free-boundary equilibrium and the fusion-reactivity birth source feed Ensemble A (10^5 markers, prompt loss) and Ensemble B (6000 markers, collisional slowing-down and wall load); the analysis stage returns the confined fraction, the wall-load map, and the arbitration verdict.

5 Discussion

5.1 Quantitative comparison with the published fast-ion literature

The result should be read against the fast-ion record. In large tokamaks, dedicated orbit-following places fusion and beam-ion confinement at the 99.9% level with first-wall loads managed to the sub-MW m^{-2} scale: the ITER energetic-ion analyses of Fasoli *et al.* [1] and Pinches *et al.* [2], and the ASCOT power-load maps of Kurki-Suonio *et al.* [8], are the reference points. Our confined fraction of 99.977% is comfortably within that band, which is the non-trivial finding: the ST orbit-width hierarchy of figure 2 ($\Delta_b/a = 0.33$, $w_p/a = 0.48$) might have been expected to degrade confinement toward the shell estimate’s 4.5%, and it does not. The reason is structural — p_φ conservation (3) holds the wide orbits inside the plasma — and is exactly the physics the spherical-torus energetic-particle review of McClements and Fredrickson [16] identifies as the reason ST fast-ion confinement is better than orbit-width scaling alone suggests. The absolute wall loads are higher *per unit loss* than in a reactor-scale device only because the wall area is small (46.4 m^2); even so, the mean loads of table 1 are two-to-four orders of magnitude below the engineering scale.

5.2 Ripple losses in context

The ripple increment — from 0.023% to 0.031% as δ_{rip} rises to 1.5% — is small because the design sits below the stochastic-diffusion threshold (13) for the bulk of the trapped population, and the modest toroidal peaking (≈ 1.4) shows that what is lost spreads over the coil period rather than forming a stripe. This is consistent with the classical Goldston–White–Boozer picture [11] and with the general finding that ripple-driven fast-ion loss is a steep function of amplitude that stays benign below threshold [1]. It also frames the one real engineering input: the actual coil-set ripple map, which this study brackets with representative amplitudes rather than computing from a specific coil geometry (see section 6).

5.3 Implications for the design point

Two design conclusions follow. First, alpha-heating retention is secure: 99.977% of the $P_\alpha = 17.01 \text{ MW}$ of birth power is deposited inside the plasma, which is the confinement premise beneath the gain $Q = 3.076$ and the integrated power balance of the platform [31]. Second, first-wall survivability against fast alphas is not a constraint: the loads of table 1 leave the plasma-facing-component lifetime to be set by the 14 MeV neutron budget [35] and by the surface thermomechanics [36], not by alpha loading. The thermal-confinement basis on which this operating point rests is independently established by the negative-triangularity turbulence-suppression result [32] and its two-code cross-validation [33], and the equilibrium on which the present orbits are followed is the same free-boundary solution used by the ideal-MHD stability assessment [34]. The fast-ion populations from auxiliary heating and current drive [37]

share the orbit topology analysed here and are confined by the same p_φ mechanism.

6 Limitations

The honest gate is one of scope. This is a guiding-centre, orbit-confinement assessment: it resolves the fast gyromotion into a drift and does not carry full-orbit finite-Larmor-radius effects at the boundary, which could matter for the small population of orbits whose turning points graze the wall. The first-wall load distribution depends on the conformal wall geometry taken from the design point, and the ripple study uses representative amplitudes rather than a specific coil-set ripple map — computing the true $\delta_{\text{rip}}(R, Z)$ from the as-designed toroidal-field coil set, and re-evaluating equation (12) locally, is the natural follow-on. Most importantly, magnetohydrodynamically driven fast-ion transport — resonant redistribution and loss by Alfvén eigenmodes and other energetic-particle modes [18, 19] — is outside the scope of an orbit-confinement calculation and is a genuinely separate question: at β_α well below the drive threshold this channel is expected to be quiescent, but it must be established by a dedicated fast-ion-MHD stability study rather than assumed. Within its stated scope, the orbit-confinement result is robust and cross-code verified.

7 Conclusion

The frozen compact spherical-tokamak breeder confines its fusion-born alphas essentially completely. Full guiding-centre orbit-following on the identical free-boundary equilibrium, run on the NVIDIA GPU HPC campaign, returns a confined fraction of 99.977% (prompt) with a first-wall alpha load of 0.083 kW m^{-2} mean and 1.94 kW m^{-2} peak, rising only to 0.033% loss and 0.121 kW m^{-2} mean under collisional slowing-down; an independent 10^6 -marker GPU follower reproduces the figure at 99.99%. This refutes the analytic large-orbit shell estimate of 4.5%, and the refutation is physical: the potato-width shell argument overcounts because it ignores conservation of the canonical toroidal momentum p_φ , which holds the wide banana and potato orbits inside the plasma despite excursions of one-third to one-half the minor radius. Finite toroidal-field ripple keeps the loss sub-percent (0.019–0.031% at 1.0–1.5% ripple) and the wall load of order 1 kW m^{-2} with only modest toroidal peaking. Alpha confinement is therefore a solved, non-limiting aspect of the Hyperion design — alpha-heating retention is secure and first-wall survivability against fast alphas is not a constraint — with the single open question, fast-ion-MHD transport, deferred to the companion stability work as a matter of scope, not of concern.

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Reproducibility. Every headline quantity in this paper is a frozen anchor in the Kronos Physics De-Risking Register [30] (doi:10.5281/zenodo.22133057): the fast-alpha orbit-width hierarchy and analytic shell estimate (serial BR-L1-A14), and the 10^5 -marker ASCOT5 orbit-following arbitration with its collisional ensemble and first-wall load map (serial BR-L2-A20). Each is regenerable from the archived FreeGS equilibrium, birth source, integrator tolerances and fixed seed (20260726) deposited with this paper, which is archived at doi:10.5281/zenodo.22645729; official version: <https://www.kronosfusionenergy.com/publications/alpha-fast-product-confinement.html>. No result depends on unpublished or proprietary data, and no economic analysis is included in the public deposit.

Data availability

The orbit-following loss records, the first-wall load map, and the figure-generation script used for figures 1–9 are archived with the deposit at doi:10.5281/zenodo.22645729 and trace to the register [30] by the serial identifiers cited inline. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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